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Third Semester- B.Tech. Course work
END SEMESTER EXAMINATION, November, 2024

ITMTC04/EIMTC04/ITMTC301/EIMTC301/ INMTC04/ INMTC301
Course title: Optimization principles and techniques

Time: 3 hours

Maximum Marks. 50

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Attempt any two parts from each question	Marks	CO
1a	Formulate the following LPP and solve it graphically. A local cafe makes two specialty smoothies: Tropical Paradise and Berry Blast. The Tropical Paradise requires 2 mangoes and 1 pineapple, while Berry Blast involves a mix of 3 different berries (strawberry, blueberry, raspberry) and half a mango. The cafe has a daily supply of 30 mangoes and 18 pineapples. Due to blender capacity and staff constraints, they can only make 25 smoothies in total per day. Each Tropical Paradise sells for Rs.8 profit, while Berry Blast sells for Rs.7 profit. How many of each smoothie should they make to maximize profit?	5	CC
1b	Consider the problem:	5	CO
1c	Solve the following LPP.	5	CO
2a	State Weak duality theorem. Use duality in solving the L.P.P. :	5	CO
2b	The following table provides all the necessary information on the availability of supply to each warehouse, the requirement of each market, and the unit transportation cost (in Rs) from each warehouse to each market.	5	CC

1b Consider the problem:

$$\text{Max } 2x_1 + x_2 + 5x_3 - 3x_4$$

subject to

$$x_1 + 2x_2 + 4x_3 - x_4 \leq 6$$
$$2x_1 + 3x_2 - 3x_3 + x_4 \leq 12$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Find a basic feasible solution with the basic variables as x_1 and x_2. Also find the optimal basic feasible solution.

1c Solve the following LPP.

$$\text{Max } 3x_1 + 2x_2 + 4x_3$$

subject to $\quad x_1 - x_2 + x_3 \geq 4, \quad 2x_1 + x_2 - x_3 \geq 12, \quad x_1, x_2, x_3 \geq 0$

2a State Weak duality theorem. Use duality in solving the L.P.P. :

$$\text{Min } z = 2x_1 - x_2 + x_3 + 5x_4 - 3x_5$$

subject to $4x_1 + 2x_2 + x_3 + x_4 = 3,$

$$2x_1 + 2x_2 + x_3 + x_5 = 2$$
$$x_j \geq 0 \quad (j = 1, 2, 3, 4, 5)$$

2b The following table provides all the necessary information on the availability of supply to each warehouse, the requirement of each market, and the unit transportation cost (in Rs) from each warehouse to each market.

	P	Q	R	S	Supply
A	6	3	5	4	22
B	5	9	2	7	15
C	5	7	8	6	8
Demand	7	12	17	9	45

The shipping clerk of the shipping agency has worked out the following schedule, based on his own experience: 12 units from A to Q, 1 unit from A to R, 9 units from A to S, 15 units from B to R, 7 units from C to P and 1 unit from C to R . (i) Check and see if the clerk has the optimal schedule. (ii) Find the optimal schedule and minimum total transport cost.

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2c	Solve the assignment problem given by the following matrix: $$\begin{array}{c\|cccc} & J_1 & J_2 & J_3 & J_4 \\ \hline M_1 & 9 & 11 & 14 & 11 \\ M_2 & 13 & 13 & 10 & 12 \\ M_3 & 14 & 13 & 12 & 13 \\ M_4 & 11 & 17 & 13 & 14 \end{array}$$ Find the optimal assignment of jobs to machines to minimize the total cost. If Machine M_3 is not available, find an alternative optimal assignment and calculate the new minimum cost.	5	CO2
3a	Write short notes on the following: (i) Saddle point of the game (ii) Mixed strategy vs pure strategy (iii) Two-person zero-sum game (iv) Minimax principle (v) Value of the game	5	CO3
3b	What are the basic assumptions behind game theory? When is the game said to be strictly determinable? Solve the following game using the graphical method. $$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 6 \\ 4 & 1 \\ 2 & 2 \\ -5 & 0 \end{bmatrix}$$	5	CO3
3c	State the relationship between game theory and linear programming. Hence, using the same or by an alternative method, solve the below payoff matrix: $$\begin{bmatrix} 3 & -1 & -3 \\ -3 & 3 & -1 \\ -4 & -3 & 3 \end{bmatrix}$$	5	CO3
4a	Solve using Gomory's method: Max Z $= x_1 - x_2$ subject to $x_1 + 2x_2 \leq 4$, $6x_1 + 2x_2 \leq 9, x_1 \geq 0, x_2 \geq 0$, x_1 and x_2 are integers	5	CO4
4b	Use Branch and Bound method to solve the ILP: $$\text{Max Z} = 7x_1 + 9x_2, \quad \text{subject to}$$ $$-x_1 + 3x_2 \leq 6, \quad 7x_1 + x_2 \leq 35, \quad x_1 \leq 7,$$ $$x_2 \leq 7, \quad x_1 \geq 0, x_2 \geq 0, \quad x_1 \text{ and } x_2 \text{ are integers}$$	5	CO4
4	Solve using Gomory's method: Max Z$= 2x_1 + x_2$, subject to $x_1 + x_2 \leq 5, 6x_1 + 2x_2 \leq 21$, $x_1 \geq 0$, $x_2 \geq 0$, x_1 is integer.The optimal simplex tableu is given below. $$\begin{array}{\|c\|c\|c\|c\|c\|\|c\|} \hline \text{Basic} & x_1 & x_2 & s_1 & s_2 & \text{RHS} \\ \hline x_1 & 1 & 0 & -1/2 & 1/4 & 11/4 \\ \hline x_2 & 0 & 1 & 3/2 & -1/4 & 9/4 \\ \hline Z & & & & & 31/4 \\ \hline \end{array}$$	5	CO4
5a	(i) Define convex and concave functions mathematically along with examples. Give geometrical description of both. (ii) For what values of a, b, c and d, the function $f(x, y) = ax^2 + 2bxy + cy^2 + d$ is convex and for what values of a, b, c and d, the function $f(x, y)$ is concave.	2+3	CO5
5b	Solve the optimization problem: $$\text{Max Z} = 7x_1^2 + 6x_1 + 5x_2 \qquad \text{subject to}$$ $$x_1 + 2x_2 \leq 10, \quad x_1 - 3x_2 \leq 9, \quad x_1 \geq 0, x_2 \geq 0.$$	5	CO5
5c	Solve: Min Z $= 6x_1^2 + 5x_2^2$ subject to $x_1 + 5x_2 = 3$, $x_1 \geq 0, x_2 \geq 0$.	5	CO5